Zihan Liang

Undergraduate AI Researcher at Xi'an Jiaotong-Liverpool University

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Research Profile

Undergraduate researcher in artificial intelligence developing reproducible machine-learning methods for mixed-frequency time-series forecasting and sequential social-media risk assessment. Experience spans Transformer and attention architectures, NLP sequence modeling, multi-horizon evaluation, imbalanced classification, and interpretability. Research communication includes an SSRN preprint and an ICSC conference poster.

Education

Xi'an Jiaotong-Liverpool University (XJTLU) — Sep 2024 – Jul 2028, expected
BEng Artificial Intelligence (Intelligent Systems) — Suzhou, China
Dual Degree: BSc (Hons), University of Liverpool, expected — Liverpool, UK

- **Academic standing:** Completed Stage 2 with a Stage 2 weighted average of 78/100 and an overall weighted average of 74/100.
- **Selected marks:** Python for AI 92, Maths for Machine Learning 86, Engineering Mathematics I 82, AI Fundamentals & Ethics 81, and Data Structures and Algorithms 78.

Research Outputs

- Q. Luo, R. Xu, **Z. Liang**, Z. Hou, and R. Zhao. *Do Macroeconomic and Geopolitical Risk Factors Improve FX Volatility Forecasts? A Mixed-Frequency Transformer Approach.* SSRN preprint, 2025. Preprint | DOI.
- **Z. Liang**, R. Xu, and R. Zhao. *Public Mental Health Monitoring: Detecting Suicide Risk in Sequential Social Media.* Conference poster presented at ICSC 2025. Code | Poster.

Research Experience

Mixed-Frequency Transformer for USD/CNH Volatility Forecasting — Aug 2025
Model Developer, XJTLU — Suzhou, China

- Implemented temporal and positional encodings for multi-horizon FX-volatility forecasting.
- Proposed cross-attention with daily technical representations as queries and lower-frequency macroeconomic and geopolitical streams as keys and values.
- Evaluated 1-, 5-, and 10-day forecasting horizons and prepared pooled and horizon-specific comparisons.
- Produced predicted-versus-actual plots, model-pipeline figures, and Integrated Gradients analysis of the 2024 U.S. election period.

Research Experience and Selected Projects

Sequential Suicide-Risk Prediction from Social Media Nov 2025
Lead Student Researcher, First Author of ICSC Poster, XJTLU Suzhou, China

- Built the RSD-15K pipeline from data acquisition and exploratory analysis through preprocessing, model training, and user-level evaluation.
- Developed a hierarchical architecture combining Twitter-RoBERTa, Bi-GRU, and attention.
- Applied emoji and token normalization, head-tail retention, and weighted focal loss for imbalanced risk categories.
- Achieved weighted F1 0.46 versus 0.43 for a RoBERTa baseline under five-fold user-level evaluation.

The Fairness Paradox: A Stratified Hybrid-Merit Model Jan 2026
Team Member, Mathematical Contest in Modeling COMAP

- The team submission examined fairness in competitions that combine judges' scores with audience votes.
- The team submission combined Softmax-based inverse inference, counterfactual comparison of rank- and percentage-based scoring, and Kruskal–Wallis analysis.
- The team proposed a Stratified Hybrid-Merit Model for competition scoring.
- Named member of a 2026 MCM Finalist team.

Grouped NIPT Timing Modelling and Fetal-Abnormality Classification Sep 2025
Statistical Modeling Contributor, CUMCM 2025 Jiangsu, China

- Studied grouped NIPT testing-time selection and fetal-abnormality classification for CUMCM 2025 Problem C.
- Contributed GLM/GAM analysis of BMI and gestational-age covariates and a BMI-grouped testing-time recommendation framework.
- Contributed logistic classification with ROC/Youden thresholding for T21, T18, and T13. The team received Second Prize in the Jiangsu Division, Undergraduate Group.

Au_{20} Scientific Visualization Support Oct 2025
Visualization and Reporting Contributor, XJTLU Suzhou, China

- Built Python utilities for Au_{20} atomic-cluster structures and bond diagrams. Code.
- Rendered animation frames in Python and assembled them into LaTeX/PDF presentation sequences.

Home Repair Assistant Robotics Jun 2025 – Aug 2025
Engineering Contributor, XJTLU SURF Suzhou, China

- Configured the ROS 2 runtime and verified node-level execution across project modules.
- Deployed and validated the codebase on NVIDIA Jetson Orin Nano for the project demonstration.

Selected Honors and Awards

Finalist 2026
Mathematical Contest in Modeling — The Fairness Paradox: A Stratified Hybrid-Merit Model

Jiangsu Provincial Second Prize 2025
Contemporary Undergraduate Mathematical Contest in Modeling — Grouped NIPT Timing Modelling

Grand Prize, Top 1 out of 50+ 2025
XJTLU Research-led Learning Symposium — Au_{20} Delta-Learning

First Prize 2025
XJTLU Research-led Learning Symposium — Grouped NIPT Timing Modelling

Second Prize 2025
XJTLU Research-led Learning Symposium — Mixed-Frequency FX Forecasting

Second Prize 2025
XJTLU Research-led Learning Symposium — Sequential Suicide-Risk Modelling

Jiangsu Provincial Third Prize 2025
"Keyou Cup" University Intelligent Robot Creativity Competition — Home Repair Assistant

Excellent Poster Award 2025
Summer Undergraduate Research Fellowship — XJTLU

Selected Presentations

International Conference on Social Computing 2025
Poster Presenter — ICSC 2025

- Presented *Public Mental Health Monitoring: Detecting Suicide Risk in Sequential Social Media*.
- Communicated the sequential modelling approach, user-level evaluation protocol, and poster findings.

XJTLU Research-led Learning Symposium Nov 2025
Poster and Oral Presenter — XJTLU

- Advanced through poster screening and delivered a 10-minute oral presentation followed by a 5-minute Q&A.
- Presented hierarchical suicide-risk modelling and received Second Prize for the project.

Technical Skills

Programming and tools: Python, Git, and LaTeX.
Python ecosystem: PyTorch, Hugging Face Transformers, scikit-learn, NumPy, Pandas, and Matplotlib.
Machine learning: Transformers, attention mechanisms, mixed-frequency and multi-horizon forecasting, cross-validation, and imbalanced classification.
NLP and data methods: RoBERTa encoding, Bi-GRU sequence modelling, statistical modelling, ROC analysis, and scientific visualization.
Languages: Chinese, native. English, English-medium degree programme with an official EMI letter available.